
Lab 9: Counter

Goals

- Introduce the use of flip-flops for sequential circuit design.
- Understand the operation of synchronous sequential circuits.

Introduction

A binary counter is a sequential circuit that increments or decrements a stored value on each clock cycle. It is widely used in digital systems for counting, timing, and generating control signals.

Lab description

Design an **8-bit synchronous counter** with the following characteristics:

- The counter must be **8 bits** wide.
- The circuit must include an **asynchronous reset signal (arstn)**.
- On each clock cycle, the counter must **increase the output cnt [7:0] by 1**.
- The circuit must include an **enable signal (en)** to control counting.
- The counter must reset when the count reaches the value specified by the **preset [7:0]** input.

Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

Video:

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.
